

Generation Interconnection System Impact Study Report

For Queue Project Q18-1077

Hamilton Peaking 230 kV

December 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1077. The Interconnection Customer proposes a 114.2 MW solar photovoltaic facility interconnecting at the Hamilton Peaking 230 kV substation in the Ohio area. The request was received on October 12, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 403 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$913,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Villa Park – Washington County Cogeneration Facility 230.0 kV	Wolfe – Barrett 500.0 kV L	101.7	1,233	19.3
Gavin Power Llc 500/345 kV Autotransformer	Gavin Power Llc – Olympus 230.0 kV Line	101.5	1,310	7.6
Villa Park – Wolfe 230.0 kV	Wolfe – Barrett 500.0 kV	101.5	1,271	19.3
Omni-Hackberry – Lucent 230.0 kV	Omni-Hackberry – Norwich 500.0 kV	100.8	823	11.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Villa Park – Washington County Cogeneration Facility 230.0 kV	Reconductor line	\$12,312,000	3% (\$363,000)
Gavin Power Llc 500/345 kV Autotransformer	Replace transformer with higher-rated unit	\$23,624,000	1% (\$262,000)
Villa Park – Wolfe 230.0 kV	Reconductor line	\$6,490,000	3% (\$191,000)
Omni-Hackberry – Lucent 230.0 kV	Reconductor line	\$6,521,000	2% (\$97,000)
	Total allocated to project		\$913,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.